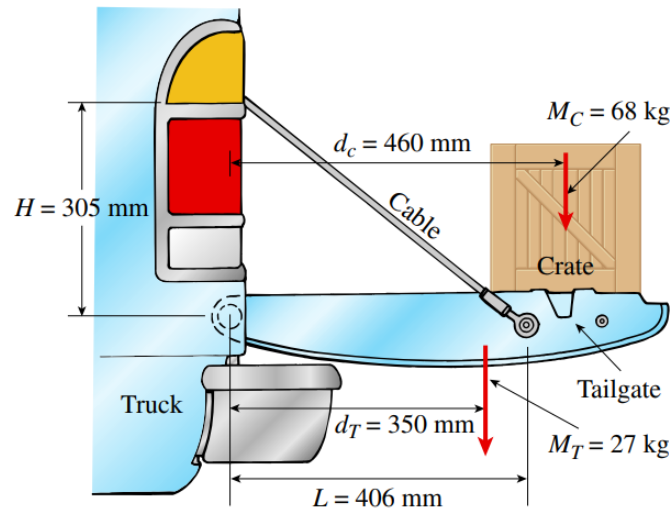


Question 1: (Review of statics)

A pickup truck tailgate supports a crate whose mass is $M_C = 68 \text{ kg}$, as shown in the figure. The mass of the tailgate is $M_T = 27 \text{ kg}$ and is supported by two cables (only one is shown in the figure). Use dimensions shown in the figure and find the tensile force T in each cable. ($g = 9.81 \text{ m/s}^2$)

**Solution:**

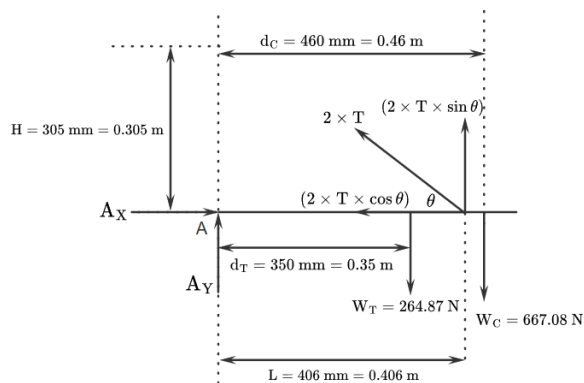
The weight of the crate: $W_C = M_C g = 68 \times 9.81 \text{ N} = 667.08 \text{ N}$

The weight of the tailgate: $W_T = M_T g = 27 \times 9.81 \text{ N} = 264.87 \text{ N}$

Suppose the tension in each cable: T , totally $2 \times T$

Assume the support reaction at hinge: A_x and A_y

Draw the FBD:



$$\tan \theta = \frac{H}{L} \Rightarrow \theta = \tan^{-1} \left(\frac{H}{L} \right)$$

$$= \tan^{-1} \left(\frac{0.305}{0.406} \right) \approx 36.915^\circ$$

According to moment equilibrium:

$$\sum M_A = 0 \Rightarrow 2T \sin \theta \times L - W_C \times d_C - W_T \times d_T = 0$$

$$\Rightarrow T = 819.258 \text{ N}$$

Question 2: (Mechanical Properties of Materials)

2.1 Following Question 1, the cable cross-sectional area is $A_e = 11 \text{ mm}^2$. What is the normal stress σ in each cable? If each cable elongates $\delta = 0.25 \text{ mm}$ due to the weight of both the crate and the tailgate, what is the Young's modulus of the cable? If Poisson's ratio of the cable is 0.3, what is the shear modulus of the cable?

Solution:

The normal stress in each cable:

$$\sigma = \frac{T}{A} = \frac{819.258 \text{ N}}{11 \times 10^{-6} \text{ m}^2} = 74.478 \text{ MPa}$$

The total length of each cable:

$$L_{\text{cable}} = \sqrt{H^2 + L^2} = 0.5078 \text{ m}$$

The normal strain in each cable:

$$\varepsilon = \frac{\delta}{L_{\text{cable}}} = \frac{0.25 \times 10^{-3} \text{ m}}{0.5078 \text{ m}} = 4.923 \times 10^{-4}$$

The Young's modulus:

$$E = \frac{\sigma}{\varepsilon} = \frac{74.478 \text{ MPa}}{4.923 \times 10^{-4}} \approx 151.286 \text{ GPa}$$

The shear modulus:

$$G = \frac{E}{2(1 + \nu)} = \frac{151.286 \text{ GPa}}{2 \times 1.3} = 58.187 \text{ GPa}$$

2.2 Please explain the differences between engineering stress and true stress. Under what circumstances would these differences be most significant?

Solution:

Both engineering stress and true stress are calculated by

$$\sigma = \frac{F}{A}, \quad \text{or,} \quad \tau = \frac{V}{A}.$$

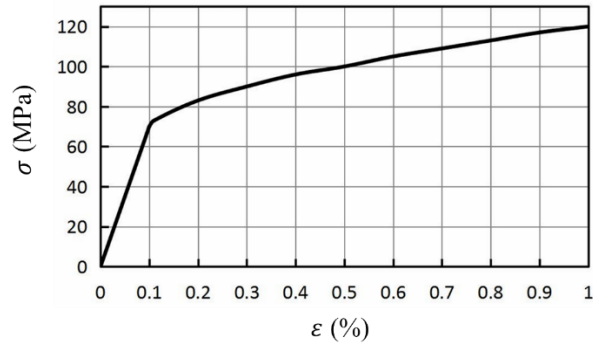
But they are based on different cross-sectional areas:

- Engineering stress – cross-sectional area in reference configuration.
(The undeformed cross-sectional area)
- True stress – cross-sectional area in current configuration.
(The deformed cross-sectional area)

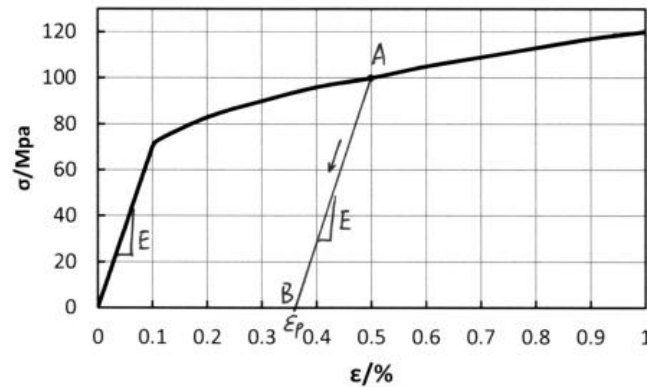
Before necking, the two cross-sectional areas are almost the same, while after necking, they deviate from each other. Therefore, after necking, these differences are most significant.

Question 3: (Tension, Compression and Shear)

Consider a straight bar having the width $b = 10.0$ mm, the thickness $h = 2.00$ mm, and the length $l = 200$ mm, which is made by a material exhibiting the stress-strain relationship as shown in the following figure. (i) Calculate the elongation of the bar when a tensile load $P = 2.00$ kN acts on the bar. (ii) Calculate the elongation of the bar after removing the load from the bar.



Solution:



(i) Stress:

$$\sigma = \frac{P}{A} = \frac{P}{bh} = \frac{2000 \text{ N}}{10 \text{ mm} \times 2 \text{ mm}} = 100 \text{ MPa}$$

From the stress-strain diagram, strain corresponds to $\sigma = 100$ MPa is

$$\varepsilon = 0.5\% = 0.5 \times 10^{-2}$$

So, the elongation

$$\lambda = \varepsilon l = 0.5 \times 10^{-2} \times 200 \text{ mm} = 1.0 \text{ mm}$$

(ii) In the unloading process, the stress decreases linearly through point A to B. The slope of AB is the Young's modulus E .

$$E = \frac{\sigma}{\varepsilon} = \frac{70 \text{ MPa}}{0.1 \times 10^{-2}} = 70 \text{ GPa}$$

When the load is removed, the strain has the value of point B as

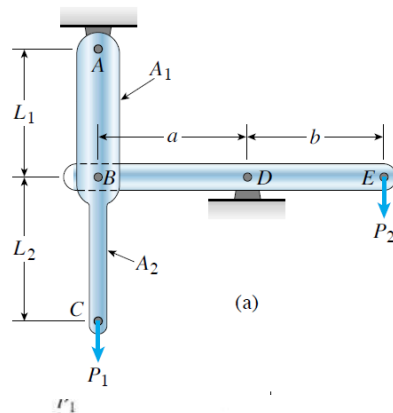
$$\varepsilon_p = \varepsilon - \frac{\sigma}{E} = 0.5 \times 10^{-2} - \frac{100 \text{ MPa}}{70 \text{ GPa}} = 0.357 \times 10^{-2}$$

The elongation:

$$\lambda = \varepsilon_p l = 0.357 \times 10^{-2} \times 200 \text{ mm} = 0.714 \text{ mm}$$

Question 4: (Axial loads and axial deformations)

4.1 A vertical steel bar ABC is pin-supported at its upper end and loaded by a force P_1 at the lower end. A horizontal beam BDE is pinned at the vertical bar at joint B and supported at point D. The beam carries a load P_2 at end E. If $P_1 = 10 \text{ kN}$, $P_2 = 25 \text{ kN}$ (disregard the weights), determine the vertical displacement at point C.



Parameters:

upper part:

$L_1 = 500 \text{ mm}$, cross-sectional area

$A_1 = 160 \text{ mm}^2$,

lower part:

$L_2 = 750 \text{ mm}$, cross-sectional area

$A_2 = 100 \text{ mm}^2$.

$E_{\text{steel}} = 200 \text{ GPa}$, $a = 700 \text{ mm}$, $b = 625 \text{ mm}$

Solution:

① P_3 at B. $M_D = 0 \rightarrow P_3 = \frac{P_2 b}{a} = 22.3 \text{ kN}$

② $R_A = P_3 - P_1 = 12.3 \text{ kN}$

③ axial force

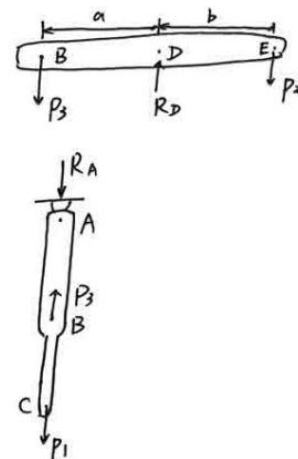
AB: $N_1 = -R_A = -12.3 \text{ kN}$ (compressive)

BC: $N_2 = P_1 = 10 \text{ kN}$ (tensile)

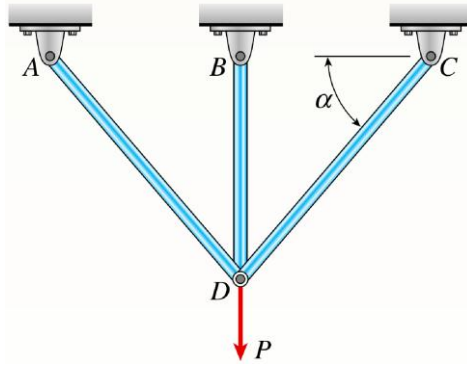
④ changes in length

$$\delta = \sum_{i=1}^2 \frac{N_i L_i}{E_i A_i} = \frac{N_1 L_1}{E_{\text{st}} A_1} + \frac{N_2 L_2}{E_{\text{st}} A_2} = -0.192 \text{ mm} + 0.375 \text{ mm} = 0.183 \text{ mm}$$

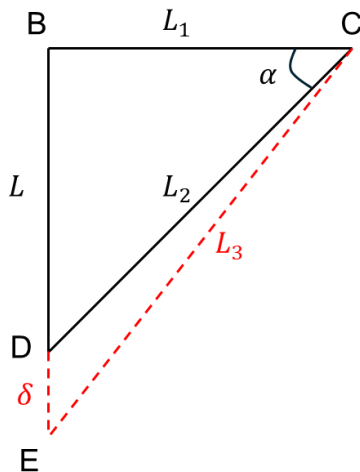
$\delta_c = \delta = 0.183 \text{ mm}$ (elongation, downward ↓)



4.2 A symmetrical framework consisting of three pin-connected bars is loaded by a force P . The angle between the inclined bars and the horizontal is $\alpha = 45^\circ$. The axial strain in the middle bar is measured as 0.2%. Determine the tensile stress in the outer bars if they are constructed of stainless steel with Young's modulus $E = 210 \text{ GPa}$.



Solution:



Suppose the length of bar BD is L , then

$$L_1 = L \cot \alpha = L$$

$$L_2 = \frac{L}{\sin \alpha} = \sqrt{2}L = 1.41421L$$

Elongation of bar BD (length DE)

$$\delta = \varepsilon_{BD}L = 0.002L$$

The new length of the outer bar CD:

$$L_3 = \sqrt{L_1^2 + (L + \delta)^2} = \sqrt{(1^2 + 1.002^2)L^2} = 1.41563L$$

The strain of bar CD:

$$\varepsilon = \frac{L_3 - L_2}{L_2} = \frac{1.41563 - 1.41421}{1.41421} = 0.1\%$$

Stress in bar CD:

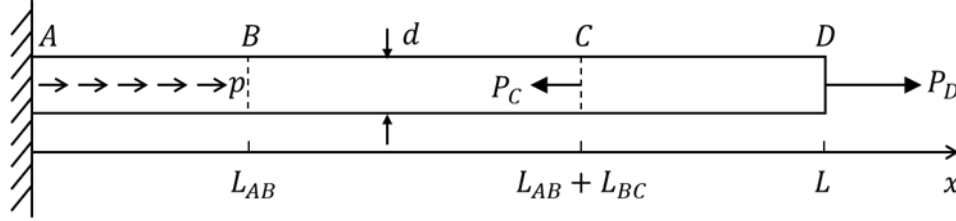
$$\sigma = E\varepsilon = 210 \text{ GPa} \times 0.001 = 210 \text{ MPa}$$

4.3 A straight prismatic circular bar AD has diameter d and length L in its undeformed state. Its left end is fixed. A uniformly distributed axial force, p per unit length, is applied on segment AB , a concentrated force, P_c , cross-section C , and a concentrated force, P_D , on the right end D . The bar undergoes linear elastic deformation with a homogeneous Young's modulus, E . The original lengths of segments AB and BC are L_{AB} and L_{BC} ,

respectively.

(i) Describe the stress/strain states in the bar as piecewise functions of x .

(ii) Calculate the elongations of segments AB , BC , CD , and the total elongation of bar AD .



Solution:

a) The bar is subjected to a reaction force, R_A , acting at the left end A in the $+x$ direction, which can be determined using the force equilibrium equation for AD:

$$R_A + pL_{AB} - P_C + P_D = 0 \rightarrow R_A = -pL_{AB} + P_C - P_D. \quad (1)$$

b) Segment AB undergoes nonuniform axial deformation. The internal force at cross-section x in Segment AB is obtained from the force equilibrium of the segment between A and x (and applying Eq. 1):

$$F(x) = -R_A - px = p(L_{AB} - x) - P_C + P_D.$$

The internal force at cross-section B is

$$F(L_{AB}) = -P_C + P_D.$$

The normal stress at cross-section x in Segment AB is

$$\sigma(x) = \frac{F(x)}{A} = \frac{p(L_{AB} - x) - P_C + P_D}{A}.$$

The normal strain at cross-section x in Segment AB is

$$\epsilon(x) = \frac{\sigma(x)}{E} = \frac{p(L_{AB} - x) - P_C + P_D}{EA}.$$

The axial displacement at cross-section x in Segment AB is

$$u(x) = u(0) + \int_0^x \epsilon(\alpha) d\alpha = 0 + \frac{(pL_{AB} - P_C + P_D)x - \frac{p}{2}x^2}{EA} = \frac{(pL_{AB} - P_C + P_D)x - \frac{p}{2}x^2}{EA}.$$

The axial displacement at cross-section B is

$$u_B = u(L_{AB}) = \frac{(\frac{p}{2}L_{AB} - P_C + P_D)L_{AB}}{EA}.$$

c) Segment BC undergoes uniform axial deformation. The internal force at cross-section x in Segment BC is obtained from the force equilibrium of the segment between A and x (and applying Eq. 1):

$$F(x) = -R_A - pL_{AB} = -P_c + P_D,$$

which is the same as the internal force at B, $F(L_{AB})$.

The normal stress at cross-section x in Segment BC is

$$\sigma(x) = \frac{F(x)}{A} = \frac{-P_c + P_D}{A}.$$

The normal strain at cross-section x in Segment BC is

$$\epsilon(x) = \frac{\sigma(x)}{E} = \frac{-P_c + P_D}{EA}.$$

The axial displacement at cross-section x in Segment BC is

$$u(x) = u_B + \int_{L_{AB}}^x \epsilon(\alpha) d\alpha = \frac{\left(\frac{p}{2}L_{AB} - P_c + P_D\right)L_{AB}}{EA} + \frac{-P_c + P_D}{EA}(x - L_{AB}).$$

The axial displacement at cross-section C is

$$u_C = u(L_{AB} + L_{BC}) = \frac{\left(\frac{p}{2}L_{AB} - P_c + P_D\right)L_{AB} + (-P_c + P_D)L_{BC}}{EA}.$$

c) Segment CD undergoes uniform axial deformation. The internal force at cross-section x in Segment CD is obtained from the force equilibrium of the segment between A and x (and applying Eq. 1):

$$F(x) = -R_A - pL_{AB} + P_c = P_D.$$

The normal stress at cross-section x in Segment CD is

$$\sigma(x) = \frac{F(x)}{A} = \frac{P_D}{A}.$$

The normal strain at cross-section x in Segment CD is

$$\epsilon(x) = \frac{\sigma(x)}{E} = \frac{P_D}{EA}.$$

The axial displacement at cross-section x in Segment CD is

$$u(x) = u_C + \int_{L_{AB} + L_{BC}}^x \epsilon(\alpha) d\alpha = \frac{\left(\frac{p}{2}L_{AB} - P_c + P_D\right)L_{AB} + (-P_c + P_D)L_{BC}}{EA} + \frac{P_D}{EA}(x - L_{AB} - L_{BC}).$$

The axial displacement at cross-section D is

$$u_D = u(L) = \frac{\left(\frac{p}{2}L_{AB} - P_c + P_D\right)L_{AB} + (-P_c + P_D)L_{BC} + P_D(L - L_{AB} - L_{BC})}{EA}.$$

d) In summary: the stress distribution is

$$\sigma(x) = \begin{cases} \frac{p(L_{AB} - x) - P_c + P_D}{A}, & 0 < x < L_{AB} \\ \frac{-P_c + P_D}{A}, & L_{AB} < x < L_{AB} + L_{BC}, \text{ where } A = \frac{\pi}{4}d^2. \\ \frac{P_D}{A}, & L_{AB} + L_{BC} < x < L \end{cases}$$

The strain distribution is

$$\epsilon(x) = \begin{cases} \frac{p(L_{AB}-x)-P_C+P_D}{EA}, & 0 < x < L_{AB} \\ \frac{-P_C+P_D}{EA}, & L_{AB} < x < L_{AB} + L_{BC}, \text{ where } A = \frac{\pi}{4}d^2. \\ \frac{P_D}{EA}, & L_{AB} + L_{BC} < x < L \end{cases}$$

The elongation of segments AB is

$$e_{AB} = u_B - u_A = u_B = \frac{\left(\frac{p}{2}L_{AB}-P_C+P_D\right)L_{AB}}{EA}. \quad (2)$$

The elongation of segments BC is

$$e_{BC} = u_C - u_B = \frac{(-P_C+P_D)L_{BC}}{EA}. \quad (3)$$

The elongation of segments CD is

$$e_{CD} = u_D - u_C = \frac{P_D(L-L_{AB}-L_{BC})}{EA}.$$

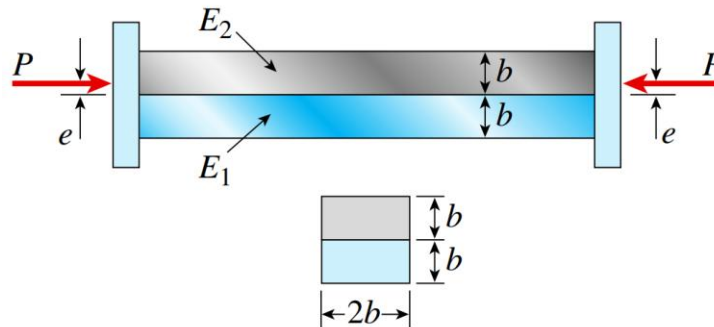
The total elongation of bar AD is

$$e_{AD} = u_D - u_A = u_D = \frac{\left(\frac{p}{2}L_{AB}-P_C+P_D\right)L_{AB}+(-P_C+P_D)L_{BC}+P_D(L-L_{AB}-L_{BC})}{EA}, \text{ where } A = \frac{\pi}{4}d^2.$$

Question 5: (Statically Indeterminate Problems)

A bimetallic bar (or composite bar) of square cross section with dimensions $2b \times 2b$ is constructed of two different metals having moduli of elasticity E_1 and E_2 (see figure, $E_2 > E_1$). The two parts of the bar have the same cross-sectional dimensions. The bar is compressed by forces P acting through rigid end plates. The line of action of the loads has an eccentricity e of such magnitude that each part of the bar is stressed uniformly in compression.

- Determine the axial forces P_1 and P_2 in the two parts of the bar.
- Determine the stresses σ_1 and σ_2 in the two parts of the bar.
- Determine the eccentricity e of the loads.



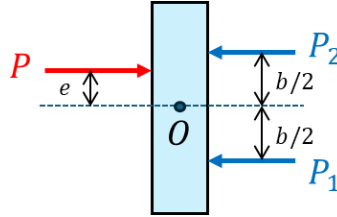
Answer:

Cross-sectional area:

$$A = 2b \times b = 2b^2$$

Assume length of the bar is L .

(i) FBD for the left rigid end plate:



Force equilibrium on the left rigid end plate:

$$\sum F_x = 0 \Rightarrow P - P_1 - P_2 = 0$$

Force-elongation relations (compressive force with magnitude P_1 and P_2):

$$\delta_1 = \frac{-P_1 L}{AE_1}$$

$$\delta_2 = \frac{-P_2 L}{AE_2}$$

Compatibility equation:

$$\delta_1 = \delta_2 \Leftrightarrow -\frac{P_1 L}{AE_1} = -\frac{P_2 L}{AE_2}$$

solved

$$P_1 = \frac{PE_1}{E_1 + E_2}$$

$$P_2 = \frac{PE_2}{E_1 + E_2}$$

(ii) The stress:

$$\sigma_1 = -\frac{P_1}{A} = -\frac{PE_1}{2b^2(E_1 + E_2)}$$

$$\sigma_2 = -\frac{P_2}{A} = -\frac{PE_2}{2b^2(E_1 + E_2)}$$

(iii) moment equilibrium at point O for the left rigid end plate:

$$\sum M_O = 0 \Rightarrow \frac{b}{2}P_2 - \frac{b}{2}P_1 - Pe = 0$$

Then we will have

$$e = \frac{b(P_2 - P_1)}{2P} = \frac{b(E_2 - E_1)}{2(E_1 + E_2)}$$